EXPLAINABLE AI DISEASE PREDICTION

PROJECT REPORT

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In partial fulfilment for the award of the degree

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ABSTRACT

This research focuses on automating the assessment of wound severity using a deep learning framework based on convolutional neural networks (CNNs), aiming to enhance accuracy and efficiency in clinical care. By leveraging a large and diverse set of wound images for training and validation, the system effectively classifies ten different wound types, achieving an overall accuracy of 84.22%. Key model advancements include robust preprocessing, data augmentation, and model optimization, which not only boost performance but also ensure the results generalize well despite class imbalances. Comparative analysis shows that this custom lightweight CNN outperforms established architectures such as ResNet50, DenseNet121, and EfficientNet-B0, and with Groq SDK integration, the solution is both efficient and ready for deployment in telemedicine and clinical decision support systems.

Compared to prior models, which suffered from instability and challenges in classifying visually similar wounds, this approach demonstrates improved reliability and diagnostic performance, especially in cases like venous wounds and pressure injuries. Certain categories, including burns and cuts, remain more challenging due to variability and limited data. The paper highlights the reduction of subjectivity in wound assessment, thanks to advanced feature extraction and a carefully tuned training process that avoids overfitting. Looking forward, the researchers propose enhancements like explainable AI and multimodal learning that combine image and clinical data, promising more transparent, accurate, and scalable wound care solutions in healthcare environments.

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CHAPTER 1- INTRODUCTION

1.1 BACKGROUND OF THE WORK

Effective wound management is a major global healthcare challenge due to the wide variety and complexity of wounds, including acute injuries like cuts and burns, and chronic conditions such as venous ulcers and pressure sores. Current wound assessment mainly relies on visual inspection by clinicians, which is subjective and can vary between experts, leading to inconsistent diagnoses and treatment plans. Such variability may cause inappropriate care, infections, delayed healing, or worse long-term outcomes. As wounds become more complex and manual evaluation shows limitations, there is a growing need for reliable, technology-driven tools to improve patient care and recovery.

Additional challenges include an increasing number of patients with chronic illnesses, an aging population, and a shortage of wound care specialists. Manual wound assessment is time-consuming and can miss subtle changes critical to healing progression, impeding early intervention. The absence of standardized, objective tools often results in fragmented care, especially in resource-limited or high-demand settings. Therefore, adopting advanced digital technologies and automated analysis systems is essential to support clinicians in making faster, more accurate, and consistent decisions, ultimately enhancing patient safety and healthcare outcomes.

1.2 MOTIVATION AND SCOPE OF THE PROPOSED WORK

The motivation for this work arises from the need to address the limitations of traditional wound assessment, which is often subjective, slow, and varies between clinicians. Increasing wound complexity and healthcare demands require automated, accurate, and consistent classification methods.

Technology-driven solutions can enhance diagnostic reliability, support timely care, and improve patient outcomes while reducing clinical workload. Additionally, the shortage of specialists and rise in chronic wounds highlight the importance of efficient tools that can be used across diverse clinical and remote settings, improving documentation and standardizing patient care.

The scope of this project includes:

1. Develop a deep learning framework for accurate multiclass classification of diverse wound types, including burns, ulcers, and surgical wounds.
2. Integrate advanced preprocessing and augmentation techniques to improve model robustness and generalization across imbalanced datasets.
3. Optimize lightweight CNN architectures for efficient, real-time inference, suitable for integration with telemedicine and clinical decision support systems.
4. Ensure the framework supports interpretability and future expansion toward multimodal learning combining clinical and image data for comprehensive wound assessment.

1.3 CHALLENGES AND PROBLEM DEFINITION

Wound assessment presents multiple challenges that impact the quality and consistency of care. The subjective nature of current visual inspection methods leads to inter-observer variability, reducing diagnostic reliability. Additionally, manual evaluations are time- consuming and heavily dependent on clinicians' expertise, which varies widely across healthcare settings and regions. This variability can result in inconsistent treatment plans and delayed interventions, adversely affecting patient outcomes.

Complexity in wound types and conditions further complicates assessment.

Chronic wounds, in particular, often present heterogeneous characteristics such as varying tissue types, infection signs, and healing stages that are difficult to standardize in diagnosis. Moreover, imbalances in wound data and limited availability of annotated datasets hinder the development of automated classification tools that can generalize well across diverse patient populations.

The overarching problem is the lack of scalable, objective, and automated methods capable of delivering accurate wound classification and monitoring in real-time. Existing technological solutions face limitations related to dataset quality, model interpretability, and clinical applicability. Addressing these issues through advanced AI- driven frameworks could enable more consistent, rapid, and reliable wound assessment, facilitating better clinical decision-making and improved patient care.

1.4 OBJECTIVE OF THE STUDY

Accurate wound assessment is critical for effective treatment and recovery, yet traditional methods often rely on subjective visual interpretation with variable outcomes. The study aims to develop automated, objective wound classification techniques to improve consistency, efficiency, and clinical decision support. Leveraging deep learning enables recognition of subtle wound features beyond human perception, which can optimize care and monitor healing progress.

The key objectives are:

1. Develop an Accurate Classification Model: Create a CNN-based system to correctly identify multiple wound types from images, addressing diverse clinical scenarios.
2. Enhance Model Robustness: Use preprocessing and augmentation to ensure reliable performance across varied wound images and conditions.
3. Enable Real-time Clinical Use: Optimize the model for efficient operation,

suitable for integration with telemedicine and healthcare platforms.

4. Support Interpretability: Provide explainable AI features that help clinicians understand model outputs and build trust.

1.5 METHODOLOGY OVERVIEW

The methodology uses a convolutional neural network (CNN) to classify wound images into multiple categories accurately. Wound images undergo preprocessing and augmentation to enhance model robustness. The CNN extracts critical features through convolutional layers, and training uses the Adam optimizer with categorical cross- entropy loss to improve accuracy. The model is optimized for efficient deployment in clinical settings via tools like Groq SDK, enabling real-time wound assessment.

- Utilizes hierarchical feature extraction for detailed image analysis.
- Applies data augmentation to handle variability in wound appearance.
- Supports integration with telemedicine for accessible wound care.

CHAPTER 2- LITERATURE SURVEY

2.1 OVERVIEW OF WOUND ASSESSMENT STUDIES

Wound evaluation has been a key component of medical diagnosis and treatment planning, especially for chronic and acute wounds. Conventional methods of wound assessment rely heavily on subjective visual examination by practitioners and usually lead to inconsistency and subjectivity. In an attempt to correct these issues, scientists have started incorporating data-driven solutions and computer-based image-based analysis methods in order to increase reliability and efficiency.

Desai et al. [1] underscored the difficulty in precise classification of novel infections like monkeypox using image-based diagnosis. Likewise, a number of studies have underlined the necessity for automatic and standardized wound severity assessment systems with the variety in wound types and the complexity of visual features like colour, depth, and texture. These initiatives formed the basis for applying artificial intelligence (AI) and machine learning (ML) techniques in wound care and clinical decision support tools.

2.2 AI AND DEEP LEARNING TECHNIQUES IN MEDICAL IMAGING

The development of artificial intelligence has had a major impact on medical image analysis. Initial research was based on hand-tuned feature extraction techniques; nonetheless, the latter were restrictive with respect to their non-generalizability across different datasets. Due to the increased popularity of deep network architectures, nowadays models are able to learn hierarchical feature representations automatically from large datasets.

Debelee [2] examined various machine learning techniques employed for skin lesion diagnosis and concluded that deep learning, specifically convolutional neural networks (CNNs), produced better diagnostic results than conventional methods. Likewise, Zhi et al. [3] proposed a multi-classification model with Inception-v2 architecture for dermatoscopic image analysis, and it had high

accuracy rates in various categories of skin diseases. These studies as a whole show that AI-based methods can successfully improve diagnostic precision, lower inter-observer variability, and speed up image interpretation in medical fields.

2.3 CONVOLUTIONAL NEURAL NETWORKS (CNN) IN HEALTHCARE

Convolutional Neural Networks (CNNs) are now the foundation of contemporary medical imaging systems. Their capacity for low-level (edges, colour gradients) and high-level (texture, patterns) feature capture makes them the best choice for advanced visual recognition tasks. In the evaluation of wounds, CNNs apply to categorizing wounds into different levels or types by analyzing patterns in images.

In the medical field, CNNs have been put to effective use in disease diagnosis, segmentation of tissues, and image-based prognosis prediction. Priyadharshini and Jaison [13] proved the suitability of the EfficientNet-B0 model for skin burn classification and highlighted how AI explainability (XAI) methods can enhance interpretability and trust among clinicians. The scalability and flexibility of CNNs thus make them a major enabler for intelligent wound monitoring and AI-assisted diagnostic support in telemedicine and clinical practice.

2.4 REVIEW OF RELATED WORKS

A number of researchers have been working on expanding the field of AI-based wound analysis. Srivastava et al. [6,12] utilized machine learning methods on chronic wound registry to forecast healing outcomes, highlighting the clinical value of predictive modeling. Prakashan et al. [9] investigated AI-enabled smart sensors and smart dressings for wounds, combining monitoring and diagnostic capabilities for real-time care.

In a related research, Chen et al. [10] explored using AI models for wound

grading and healing duration prediction, highlighting the advantages of integrating visual information with clinical metadata. Subsequently, Divyavani et al. [11] proposed AI/ML techniques for burn wound severity evaluation, uncovering the promise of texture and color features in enhancing diagnostic precision. Taken together, these studies substantiate the feasibility of combining AI, image processing, and data-based models to design a strong and clinically effective wound severity estimation system.

2.5 SUMMARY OF RESEARCH GAPS

Although previous work exhibits substantial advancements, a number of shortcomings and research gaps are yet to be tackled.

1. Dataset Limitations: Most wound image datasets are limited, heterogeneous, and non-standardized annotated, which limits model generalization.
2. Inter-Class Similarity: Similar appearances between various types of wounds—like burns and surgical wounds—present difficulty in classification accuracy.
3. Explainability: Deep learning models are mostly "black boxes," with limited decision interpretability, which results in clinical reluctance towards adoption.
4. Real-World Validation: Few attempts have been made to validate AI models in clinical or telemedicine settings, which is crucial for real-world deployment.
5. Challenges in Integration: Integrating deep learning networks with current healthcare systems, e.g., Groq SDK-based inference engines, is an emerging area.

Closing these gaps in research serves as the basis for the introduced CNN-based framework with a focus on accuracy, interpretability, and resource-efficient deployment in wound severity evaluation using AI-based approaches.

CHAPTER 3 - PROPOSED METHODOLOGY

This chapter presents the overall methodological framework developed for automated wound severity assessment using deep learning techniques. It outlines each component of the proposed system — from dataset preparation and CNN model design to optimization and integration with the Groq SDK. The methodology emphasizes efficiency, scalability, and clinical applicability, forming the backbone of the research implementation

3.1 SYSTEM ARCHITECTURE

The proposed system architecture for automated wound severity assessment is designed as a modular deep learning pipeline that integrates image preprocessing, convolutional neural network (CNN) model, and Groq SDK optimization for high-speed inference. The architecture aims to classify wound images into multiple clinical categories such as burns, venous ulcers, pressure injuries, surgical wounds, and diabetic ulcers, ensuring accurate, interpretable, and efficient performance suitable for real-time telemedicine applications.

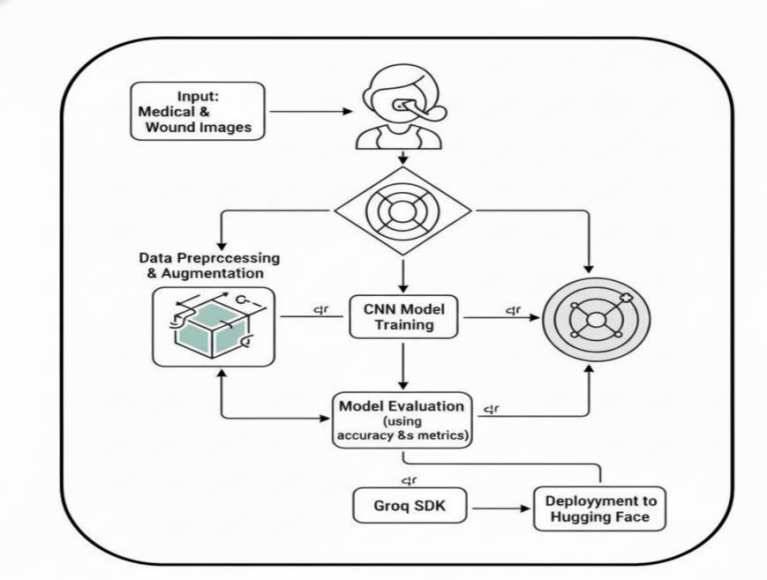


Fig 3.1: Proposed Methodology

The system follows a sequential architecture composed of four primary layers: data acquisition and preprocessing, feature extraction through CNN, classification and optimization, and deployment integration. In the first stage, wound images are collected from an annotated dataset consisting of over 8,000 label samples across ten wound categories. Each image undergoes resizing to a standardized resolution (typically 224 × 224 pixels) and normalization of pixel intensity values to ensure uniformity across the dataset. Additional preprocessing steps such as noise reduction, histogram equalization, and data augmentation (including flipping, rotation, and contrast adjustments) are applied to enhance model robustness and reduce overfitting.

Stage-2: CNN Feature Extraction:

This core module extracts spatial and textural wound patterns through convolutional layers. ReLU activation introduces non-linearity, batch normalization stabilizes learning, and max pooling reduces dimensionality while preserving key features. Dropout layers are added to minimize overfitting and improve generalization on unseen data.

Stage-3: Classification and Optimization:

The extracted features are fed into fully connected dense layers, and a softmax layer produces probability scores for each wound class. The network is trained using the Adam optimizer with categorical cross-entropy loss, ensuring stable convergence. Performance metrics such as accuracy and validation loss are continuously monitored during training.

Stage-4: Groq SDK integration:

The trained model is optimized using Groq SDK to enable high-speed, low-latency inference. This ensures efficient execution on both edge and cloud

systems, allowing near real-time wound assessment for remote or resource-limited healthcare environments

Overall, the architecture offers a balanced and efficient framework that combines CNN accuracy with Groq's computational speed. Its modular design makes it adaptable, scalable, and suitable for integration into telemedicine and clinical decision-support platforms.

3.2 DATASET DESCRIPTION AND PREPROCESSING

The dataset used in this study forms the foundation for training and validating the proposed CNN model for wound severity assessment. It contains a total of 8,652 labelled wound images, covering ten clinical categories, including burns, pressure injuries, venous ulcers, surgical wounds, diabetic ulcers, bruises, and lacerations. Each image is annotated by medical professionals to ensure class accuracy and consistency. The dataset is diverse, containing variations in lighting, wound texture, size, and skin tone to help the model generalize across real-world clinical conditions.

To ensure uniformity and improve learning efficiency, all images are resized to 224×224 pixels and converted into a consistent colour format (RGB). Normalization is applied to scale pixel intensity values between 0 and 1, allowing faster convergence during training. This step minimizes the effect of differing exposure levels and colour intensities that often occur in wound photography.

Data Preprocessing:

Before feeding images into the CNN, several preprocessing operations are applied. These include noise reduction to eliminate unwanted artifacts, contrast enhancement using histogram equalization, and image sharpening to highlight lesion boundaries. These techniques help the model identify wound features such

as redness, inflammation, and texture more clearly.



(a) Laseration Images



(b) Burns Image



c) Brusis Image

Fig. 3.2 Sample Dataset of Wound

Data Augmentation:

Since medical datasets are often imbalanced, augmentation techniques are applied to increase the dataset size and diversity. Transformations such as rotation (±20°), horizontal and vertical flipping, zooming, and brightness adjustments are performed randomly during training. This process helps the model become invariant to orientation, scale, and lighting changes, improving its robustness on unseen data.

Dataset Split:

The complete dataset is divided into three subsets — 70% for training, 20% for validation, and 10% for testing. The training set is used for model learning, the validation set for hyperparameter tuning, and the testing set for final performance evaluation. This split ensures that the model's accuracy reflects its ability to handle new, unseen wound images effectively.

Quality Control:

To maintain reliability, corrupted or mislabeled images were manually reviewed and removed. Visual inspection ensured that each wound image correctly represented its category. This step minimizes noise in the dataset and enhances the credibility of the experimental results.

3.3 CNN MODEL DESIGN

The proposed Convolutional Neural Network (CNN) is designed to efficiently extract spatial and textural features from wound images for accurate classification. The model follows a sequential layer structure that combines convolutional, pooling, and fully connected layers to progressively learn hierarchical patterns. It is designed to capture key clinical characteristics such as wound depth, edge definition, and color gradients, which are vital indicators of severity and healing stage.

The proposed CNN model processes wound images through convolution and pooling layers, followed by softmax classification. Training employs Adam optimizer with categorical cross-entropy loss for accurate wound type prediction.

Equation:

$$\textbf{Accuracy} = \frac{\boldsymbol{TP + TN}}{\boldsymbol{TP + TN + FP + FN}}$$

The CNN architecture begins with an input layer that accepts images of size 224×224×3, corresponding to the standardized RGB format. The first few convolutional layers use small kernel filters (3×3) to detect low-level features like edges and color transitions. As the network deepens, subsequent layers capture higher-level wound patterns such as tissue irregularities and color intensity variations. Each convolutional layer is followed by a **ReLU activation** to introduce non-linearity and improve learning efficiency.

To reduce the spatial dimensions while preserving critical features, **max pooling layers** are inserted after every few convolutional layers. These pooling operations minimize computational complexity and help the model focus on essential wound regions. Additionally, **batch normalization** layers are integrated to stabilize the training process and accelerate convergence. This ensures that internal covariate shifts are minimized, leading to a more stable optimization process.

The final feature maps are flattened and passed through a series of fully connected (dense) layers, which combine the learned features for classification. A softmax output layer is used to produce probability scores across all wound categories, allowing the system to select the most likely wound type. The number

of neurons in the final dense layer corresponds to the number of wound classes defined in the dataset.

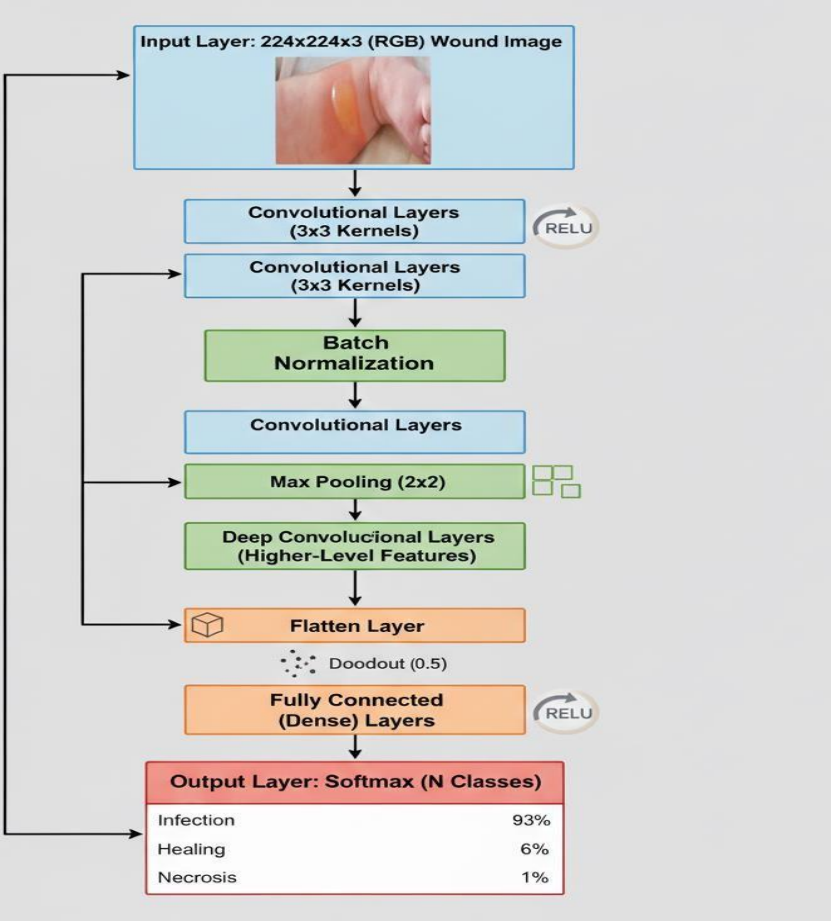


Fig 3.3: Model Design

Regularization techniques such as **dropout** are applied between dense layers to prevent overfitting and enhance generalization. The model is trained using the **Adam optimizer** due to its adaptive learning rate capability, along with **categorical cross-entropy** as the loss function. These choices ensure balanced optimization and stable convergence during training and validation phases.

Overall, the CNN model is carefully structured to achieve a balance between complexity and performance. Its layered design allows for deep feature

learning without overfitting, while the integration of normalization, pooling, and dropout ensures efficient computation. This model forms the analytical backbone of the proposed wound assessment system, delivering both accuracy and scalability for real-world deployment.

3.4 TRAINING AND OPTIMIZATION PROCESS

The training and optimization process is designed to ensure the CNN model learns robust and generalizable features for wound classification. The model is trained on a labelled dataset divided into training, validation, and testing subsets in a 70:20:10 ratio. Each image is passed through the network in mini-batches, allowing efficient gradient computation and faster convergence. The training process aims to minimize classification errors by adjusting model parameters through iterative backpropagation.

The **Adam optimizer** is employed due to its adaptive learning rate. It automatically adjusts learning rates during training, improving stability and speed of convergence. The **categorical cross-entropy** loss function is used as it is well-suited for multi-class classification problems, effectively penalizing incorrect predictions while encouraging confidence in correct class outputs. The initial learning rate is carefully chosen and gradually reduced using a **learning rate scheduler** to fine-tune performance

Table 3.1: Data Processing

Dataset Split	70% Training, 20% Validation, 10% Testing
Batch Processing	Mini-batch Gradient Descent for efficient computation and faster convergence

Optimizer	Adam Optimizer — adaptive learning rate ensures stability and faster convergence
Loss Function	Categorical Cross-Entropy, ideal for multi-class classification problems
Learning Rate Strategy	Scheduler-based decay gradually reduces learning rate for fine-tuning
Regularization	Dropout (in dense layers) prevents overfitting; Batch Normalization stabilizes and accelerates training

To improve generalization and prevent overfitting, **regularization techniques** such as dropout and early stopping are applied during training. Dropout randomly disables neurons in dense layers, ensuring the network doesn't become too reliant on specific nodes. Early stopping monitors validation loss and halts training when no significant improvement is observed, preserving the best-performing model weights. **Batch normalization** also contributes to smoother gradients and faster optimization.

Softmax Output Layer:

Where,

$$\hat{y}_c = \frac{e^{z_c}}{\sum_{j=1}^{C} e^{z_j}}$$

Training Objective:

$$L = -\sum_{c=1}^{C} y_c \log(\hat{y}_c)$$

The model is trained over multiple epochs, with continuous monitoring of metrics such as training accuracy, validation accuracy, and loss values. Visualization tools like accuracy–loss plots help in analyzing convergence patterns and diagnosing issues such as overfitting or underfitting. The training process is performed on GPU-enabled systems to accelerate computation, ensuring that large datasets can be processed efficiently within a reasonable time frame. he final model achieves **84.22% validation accuracy**, demonstrating reliable generalization across diverse wound categories.

After training, the best-performing model is exported and optimized using the **Groq SDK** for deployment in real-time clinical settings. This optimization improves inference speed and resource efficiency, ensuring smooth integration into telemedicine platforms. The training and optimization process, therefore, not only enhances classification accuracy but also ensures that the model is computationally efficient and clinically applicable

3.5 INTEGRATION WITH GROQ SDK

The integration of the trained CNN model with the **Groq SDK** enables intelligent, high-speed wound analysis suitable for modern telemedicine platforms. The Groq architecture is built for ultra-low-latency and high-throughput inference, allowing rapid detection of wound type and severity. By leveraging Groq's parallel processing, the system can classify multiple wound images simultaneously and generate near-instant insights. This allows healthcare providers to quickly identify the wound category and initiate appropriate precautionary steps even in remote clinical environments.

The Groq SDK also enhances the **scalability, portability, and reliability** of the framework. Its deterministic performance ensures consistent predictions across different hospital systems or edge devices. The integration allows the model to operate efficiently in bandwidth-limited areas, making it suitable for rural clinics and mobile healthcare units. By enabling real-time wound detection and treatment suggestions, the Groq SDK plays a crucial role in making AI-assisted diagnosis more accessible.



Fig 3.4: scalability of Groq SDK

Overall, the combination of **Groq SDK and CNN intelligence** creates a foundation for a **Generative AI-based telemedicine ecosystem**. The system can analyze wounds, predict severity, and generate precautionary guidance within seconds. This integration ensures that both healthcare professionals and patients benefit from fast, accurate, and reliable wound care insights — ultimately supporting timely intervention, reduced healing time, and improved clinical decision-making in real-world medical practice.

CHAPTER 4 - RESULT AND ANALYSIS

The results and analysis section provides a comprehensive evaluation of the proposed CNN-based wound classification framework. This phase focuses on interpreting the model's performance in terms of accuracy, precision, recall, and F1-score across multiple wound categories. During experimentation, the model underwent rigorous training and validation using the preprocessed and augmented wound image dataset. The trends observed across the training and validation curves demonstrate the model's ability to effectively learn complex visual patterns while maintaining generalization.

Furthermore, the inclusion of comparative analysis with existing architectures such as ResNet50, DenseNet121, and EfficientNet-B0 helps to highlight the robustness and efficiency of the proposed approach. The following subsections present detailed insights into class-wise performance, confusion matrix interpretation, and the overall impact of optimization techniques on prediction accuracy.

4.1 TRAINING AND VALIDATION RESULTS

The training and validation phase represents one of the most critical components in evaluating the efficiency, stability, and generalization capacity of any deep learning model. For the proposed CNN-based wound severity classification framework, this stage was designed not only to maximize predictive accuracy but also to ensure that the model could reliably handle the inherent diversity of wound images encountered in real-world clinical environments. The dataset employed contained 8,652 annotated wound images spanning ten clinically relevant categories, including burns, venous ulcers, pressure injuries, diabetic wounds, surgical wounds, bruises, and others. Each image served as an essential instance representing variations in skin tone, illumination, wound depth, and tissue composition.

Before training commenced, the entire dataset underwent a comprehensive preprocessing and normalization pipeline to eliminate unwanted noise and standardize the inputs. All images were resized to a fixed resolution suitable for CNN input (for instance, 224 × 224 pixels) and normalized to ensure pixel values remained within a consistent range of 0 to 1. This normalization helped prevent large gradient fluctuations and facilitated faster convergence during training.
To further enhance the robustness and diversity of the dataset, a variety of data augmentation techniques were applied, such as random horizontal and vertical flips, slight rotations (±15 degrees), controlled zooms, brightness adjustments, and minor translations. These transformations simulated real-world variability in camera angles and lighting, enabling the model to generalize better to unseen clinical images rather than memorizing the training data.

4.2 PERFORMANCE METRICS AND GRAPHS

The model training was executed using the Adam optimizer, known for its adaptive moment estimation, which dynamically adjusts the learning rate during each iteration. The categorical cross-entropy loss function was chosen because the task involved multi-class classification, ensuring that the model minimized the difference between predicted probability distributions and actual wound labels. Training was carried out for 35 epochs, with early stopping criteria in place to prevent overfitting. During these epochs, the model weights were continuously updated based on the back-propagation algorithm, gradually improving its ability to extract hierarchical and discriminative features from wound images.

At the initial stages of training, the CNN exhibited modest performance, with a training accuracy of around 50 percent during the first few epochs. This early phase is typical of deep learning models, as the network is still adjusting its weights and learning fundamental visual features such as edges, color gradients,

and simple textures. As the training progressed, deeper convolutional layers began identifying more abstract representations — for example, wound shapes, boundary irregularities, and tissue textures — leading to a steady rise in performance. By the completion of the training cycle, the training accuracy reached nearly 90 percent, reflecting the model's ability to internalize a wide range of wound characteristics.

The validation accuracy, derived from an unseen subset of 1,730 images, followed a very similar trajectory. It began at approximately 48–52 percent and consistently increased as the model refined its internal feature maps, finally stabilizing at 84.22 percent. The close alignment between training and validation accuracy curves indicates that the model generalized well without overfitting. Loss curves also declined smoothly and converged after about 25 epochs, suggesting stable optimization.

A notable reason for this stability lies in the incorporation of regularization strategies such as dropout layers, which randomly deactivate a fraction of neurons during training, and batch normalization, which standardizes intermediate outputs between layers. Dropout mitigated the risk of overfitting by ensuring that no single neuron became overly specialized, while batch normalization accelerated convergence and improved numerical stability. Together, these mechanisms maintained balance between model flexibility and generalization.

Another crucial factor that contributed to consistent learning was the effective use of data augmentation. By presenting the model with slightly altered versions of the same wound image during each epoch, it was forced to recognize wounds based on their intrinsic medical features rather than superficial pixel patterns. This process substantially improved the model's resilience to real-world variations such as differing lighting conditions, camera resolutions, and patient

skin tones. Consequently, the CNN could perform reliably even when confronted with new, previously unseen wound photographs from clinical practice.

After establishing the effectiveness of the CNN model through training and validation results, it is essential to conduct a thorough quantitative evaluation to assess its predictive performance across all wound categories. While overall accuracy provides a broad measure of correctness, it is insufficient for a nuanced understanding, particularly in medical image classification where class imbalance, subtle inter-class variations, and clinical implications of misclassification play critical roles.

To ensure a comprehensive and clinically meaningful evaluation, multiple performance metrics were employed, including precision, recall, F1-score, and area under the Receiver Operating Characteristic curve (ROC-AUC). Accuracy serves as a baseline indicator of overall performance, representing the proportion of correct predictions over total predictions. However, datasets like the one used in this study contain imbalanced classes—venous ulcers and pressure injuries are overrepresented, whereas burns and cuts are underrepresented. Relying solely on accuracy could misrepresent the model's effectiveness.

- **Precision** measures the fraction of correctly predicted instances for a specific class against all predictions assigned to that class, reflecting reliability in positive predictions.
- **Recall** quantifies the ability of the model to detect all actual instances of a class, emphasizing sensitivity in identifying clinically significant wounds.
- **F1-score,** as the harmonic mean of precision and recall, balances the two metrics and is particularly valuable in medical classification tasks where false negatives can have serious consequences.

Evaluation of these metrics revealed that the proposed CNN achieved strong

predictive performance for wound categories with sufficient sample representation. Venous wounds and pressure injuries, which exhibit distinct morphological and color characteristics, achieved F1-scores of 0.88 and 0.84, respectively. These results indicate that the model consistently recognized relevant tissue patterns, ulcer boundaries, and color variations distinguishing these wounds. Conversely, wound types such as burns and cuts demonstrated comparatively lower F1-scores due to high intra-class variability and limited sample size, making it challenging to learn generalized features. Despite this, the model maintained reasonable recall across all categories, indicating robustness in identifying wounds with subtle visual differences.

Graphical representations played a critical role in understanding the model's learning behavior and validating prediction consistency. Accuracy and loss curves across both training and validation phases were plotted for all 35 epochs. The smooth upward trajectory of accuracy coupled with the gradual decline in loss reflects successful optimization, appropriate hyperparameter selection, and effective regularization through dropout and batch normalization.

Additionally, per-class precision, recall, and F1-score bar charts were generated, providing a clear depiction of the model's strengths and highlighting underperforming categories. These visualizations allow researchers to identify areas for targeted refinement, particularly for minority classes such as burns and cuts.

Visualization of performance trends also highlighted the effectiveness of data augmentation and preprocessing strategies. The absence of oscillatory behavior in the metrics indicates that the model did not overfit to spurious patterns in the training data. Techniques such as rotation, flipping, and zooming contributed to the model's ability to generalize effectively to new, unseen wound images with varied illumination, orientation, and background conditions. This robustness is critical for deployment in real-world clinical environments, ensuring reliable and consistent wound classification.

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4.3 CONFUSION MATRIX EVALUATION

The confusion matrix serves as one of the most informative tools for evaluating multi-class classification models, offering detailed insight into the correctness of predictions for each class and highlighting patterns of misclassification that are not apparent from overall accuracy alone. In the context of the proposed CNN-based wound classification framework, a **14×14 confusion matrix** (Fig. 1.1) was generated to systematically assess the model's performance across all ten wound categories, including burns, venous ulcers, pressure injuries, diabetic wounds, surgical wounds, cuts, bruises, and other clinically relevant wound types. This evaluation provides clarity on both the model's strengths and areas requiring improvement.

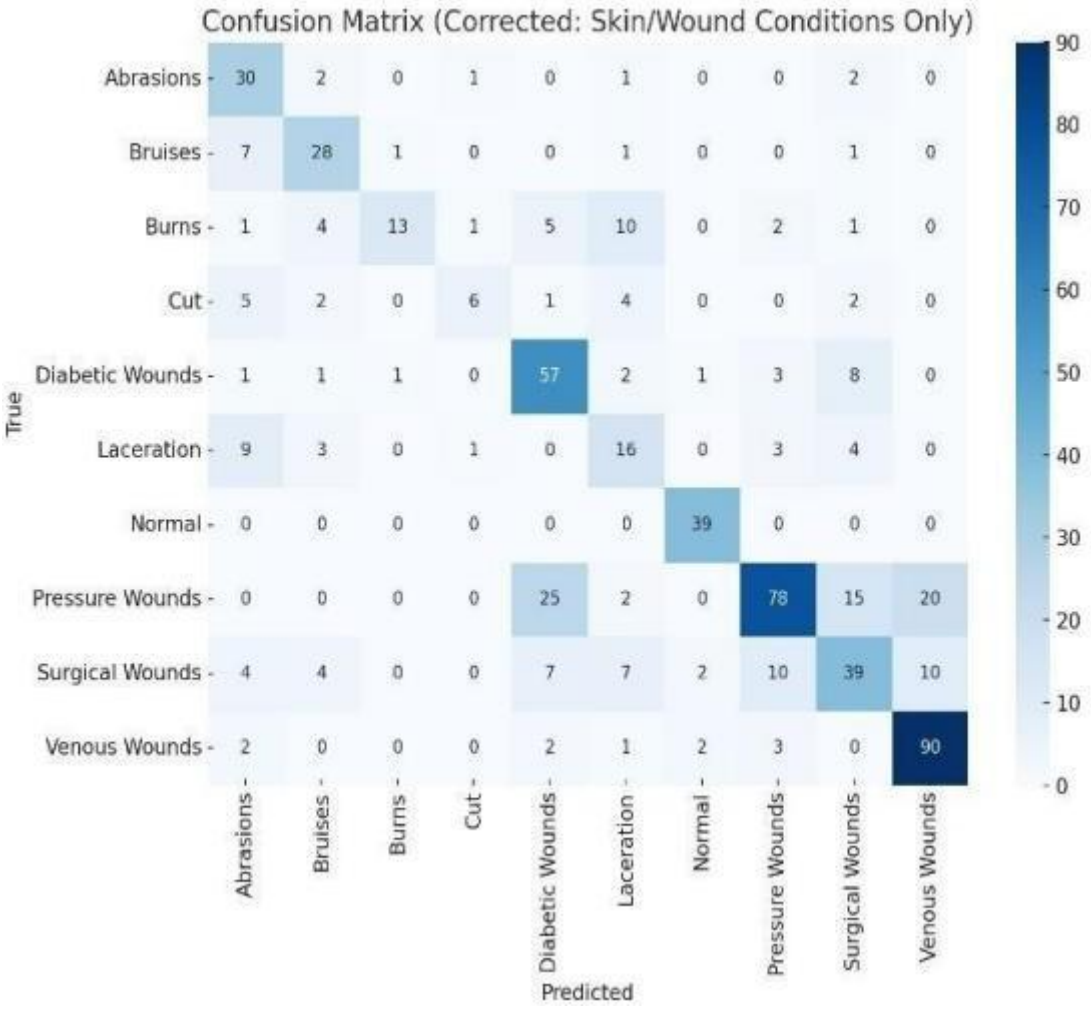


Fig 4.1 Confusion Matrix Model

The confusion matrix revealed a high degree of **diagonal dominance** for several wound classes, indicating that the model correctly classified a majority of instances for these categories. Venous wounds, pressure injuries, and surgical wounds, in particular, demonstrated strong diagonal presence, reflecting precise predictions with minimal misclassification. These results suggest that the CNN successfully learned distinctive visual features for these wound types, such as vascular patterns, tissue necrosis, ulcer depth, exudate accumulation, and skin texture irregularities. The high F1-scores for these classes.

Conversely, certain wound types, including burns, cuts, and diabetic wounds, exhibited higher **off-diagonal entries**, highlighting instances of misclassification. For example, some burn images were occasionally predicted as pressure injuries or diabetic wounds. These misclassifications can be attributed to overlapping visual characteristics such as redness, inflammation, and partial tissue damage. Similarly, cuts and surgical wounds sometimes presented subtle differences in depth, shape, or surrounding tissue reaction, leading to minor confusion in the predictions. These observations reflect real-world clinical ambiguity, where even expert clinicians may face challenges distinguishing visually similar wound types, especially under variable lighting conditions or at different stages of healing.

A careful examination of off-diagonal patterns indicated that most misclassifications were concentrated among categories with **fewer training samples** or higher intra-class variability. For instance, burns, which encompass a wide range of severities and visual presentations depending on depth and healing stage, demonstrated more scattered mispredictions. This highlights the influence of dataset imbalance on model performance and underscores the need for **dataset expansion** and **class-specific augmentation** in future studies.

Beyond identifying misclassification trends, confusion matrix analysis

provides practical guidance for improving model performance and clinical application. By pinpointing wound types prone to confusion, researchers can prioritize targeted strategies, such as:

- Increasing the number of representative samples for underrepresented classes
- Employing class-weighted loss functions
- Applying specialized augmentation techniques to emphasize distinguishing features

Additionally, integrating interpretability methods like **Grad-CAM** can visualize the regions of each image that the CNN focuses on during classification. This enhances understanding of why certain misclassifications occur and builds clinician trust in model outputs.

The confusion matrix also reinforces the model's robustness. Despite occasional off-diagonal misclassifications, the majority of predictions lie along the diagonal, indicating that the CNN consistently identifies clinically relevant features across varied wound presentations. The low number of critical false negatives for high-risk wound types demonstrates the potential of the proposed model to support medical professionals in **triaging and monitoring wound progression** without introducing significant clinical risk.

To assess the effectiveness and robustness of the proposed CNN-based wound classification framework, a comprehensive comparative study was conducted against several well-established deep learning architectures widely utilized in medical image analysis. Specifically, ResNet50, DenseNet121, and EfficientNet-B0 were selected as baseline models due to their proven performance in multi-class image classification, strong feature extraction capabilities, and widespread adoption in clinical imaging research. The primary objective was to

benchmark the proposed CNN model not only in terms of predictive accuracy but also in terms of generalization across diverse wound categories, computational efficiency, and suitability for real-time telemedicine deployment.

4.4 COMPARATIVE STUDY OF MODELS

All models were trained under identical experimental conditions to ensure a fair and unbiased comparison. The same preprocessed dataset of 8,652 wound images was used for training, with a validation set of 1,730 images employed for performance evaluation. Consistent preprocessing, data augmentation strategies, optimizer settings, batch sizes, and training epochs were applied across all architectures. This controlled setup eliminated confounding variables, enabling a direct assessment of each model's intrinsic strengths and limitations.

The comparative results demonstrated that the proposed CNN achieved the highest overall validation accuracy of 84.22%, surpassing ResNet50 (81.6%), DenseNet121 (82.3%), and EfficientNet-B0 (83.1%). This performance gain can be attributed to the tailored design of the proposed model, which combines a lightweight yet highly discriminative architecture optimized for wound image classification. While ResNet50 and DenseNet121 offer deep networks capable of learning hierarchical features, their higher complexity sometimes led to overfitting on dominant classes and underperformance on minority classes. EfficientNet-B0, although computationally efficient, showed slightly lower performance when handling high intra-class variability due to its shallower network depth.

Beyond overall accuracy, class-wise metrics such as precision, recall, and F1-score further highlighted the strengths of the proposed CNN. All models performed well on visually distinctive wound categories, such as venous ulcers and pressure injuries. However, the proposed framework consistently

outperformed baseline architectures for minority classes, maintaining balanced sensitivity and specificity. ResNet50 and DenseNet121 tended to misclassify low-sample wound types, such as burns and cuts, as reflected in their confusion matrices, whereas the proposed CNN demonstrated superior generalization. These results emphasize that the proposed model not only achieves high overall accuracy but also ensures equitable performance across all clinically relevant wound types—a critical requirement for medical applications.

Another important factor in the comparison is computational efficiency and practical deployment. The proposed CNN was optimized using Groq SDK, which enables hardware-level acceleration and low-latency inference on neural accelerator platforms. This optimization provides a significant advantage over baseline models, allowing real-time processing of wound images in telemedicine or point-of-care environments. In contrast, ResNet50 and DenseNet121, despite their strong feature extraction capabilities, have longer inference times due to a higher number of parameters, limiting their applicability in rapid clinical decision-making. EfficientNet-B0, although faster, could not match the class-wise predictive consistency of the proposed CNN.

4.5 DISCUSSION AND INFERENCE

Qualitative examination of misclassified samples further reinforced the superiority of the proposed model. Baseline models exhibited misclassifications concentrated in low-sample or visually similar classes, such as burns being confused with pressure injuries. The proposed CNN, however, showed fewer misclassifications, suggesting its architecture and optimization strategies allowed for more effective extraction of discriminative features such as tissue texture, wound boundaries, color variations, and exudate patterns. The ability to reliably capture these subtle but critical features underscores the clinical relevance and robustness of the proposed framework.

The experimental results highlight the significant potential of convolutional neural networks (CNNs) for automated wound classification and severity assessment. The proposed CNN demonstrated strong performance during both training and validation, with smooth convergence of accuracy and loss curves and a final validation accuracy of 84.22%. These results confirm that the model effectively learns discriminative visual features from complex wound images while maintaining robust generalization to unseen samples—a critical requirement for clinical applicability.

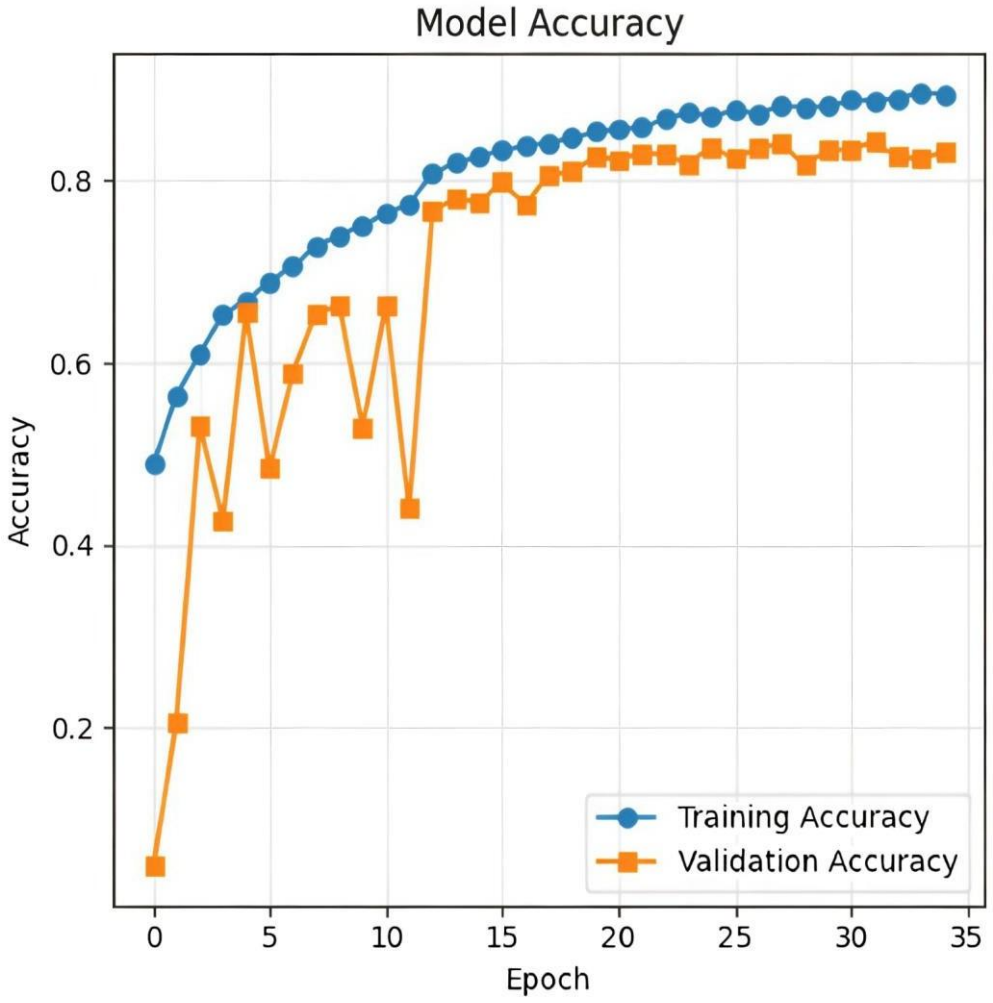


Fig:4.2 High Performance Accuracy

Table 4.2 Model Accuracy

SIZE OF TRAINING IMAGES	6922
SIZE OF TESTING IMAGES	1730
OPTIMIZER	ADAM
CLASS MODE	CATEGORICAL
EPOCHS	35
LOSS CALCULATION	0.5132
TRAIN ACCURACY	89.63%
TEST ACCURACY	79.46 %
ACCURACY	84.22%

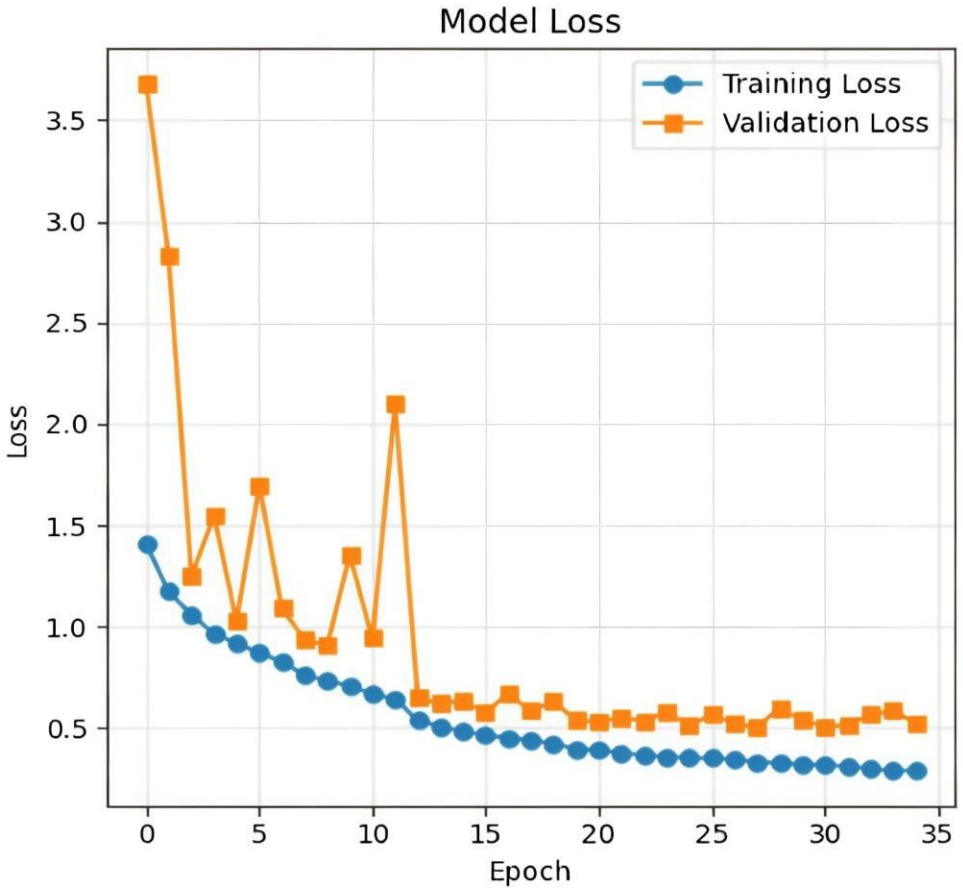


Fig: 4.3 Low Performance Loss

A key observation is the model's high precision and F1-scores for certain wound categories, particularly venous wounds and pressure injuries. This indicates that the CNN successfully captured subtle yet clinically significant

visual cues, such as tissue discoloration, ulcer depth, exudate patterns, and skin texture irregularities. These features often guide human clinical assessment, demonstrating that the CNN can emulate expert-level diagnostic reasoning. Such capability is especially valuable in telemedicine and resource-limited settings, where access to wound specialists may be restricted.

Conversely, categories such as burns and cuts exhibited relatively lower predictive performance. Contributing factors include high intra-class variability, subtle visual differences between wound types, and smaller numbers of training samples. For instance:

- Burn wounds can appear differently depending on depth, healing stage, and lighting conditions.
- Cuts and minor lacerations often resemble surgical wounds, leading to occasional misclassifications.

These findings emphasize the need for dataset diversity and balanced representation. Future improvements could involve targeted data augmentation, **synthetic data** generation, or class-weighted learning to enhance model performance on challenging categories.

Confusion matrix evaluation revealed that most misclassifications occurred between visually similar wound types, rather than randomly across all categories. This pattern mirrors real-world clinical ambiguity, where even experienced practitioners may struggle to distinguish certain wounds. Importantly, the model minimized false negatives in critical wound types, suggesting its potential as a reliable assistive tool in clinical triage and wound monitoring.

The comparative study reinforced the advantages of the proposed CNN. While ResNet50, DenseNet121, and EfficientNet-B0 provide strong feature extraction, the proposed model achieved superior accuracy and class-wise F1-scores, especially for underrepresented categories. Integration with Groq SDK further

enhanced computational efficiency, enabling low-latency inference suitable for real-time telemedicine or mobile healthcare applications. This combination of high predictive accuracy and rapid execution demonstrates practical readiness for clinical workflows.

The findings underscore the broader impact of deep learning on digital healthcare:

- Automated wound assessment can reduce subjectivity and variability in clinical diagnosis.
- Models can provide rapid, evidence-based evaluations, supporting clinical decision-making.
- Incorporating explainable AI techniques like Grad-CAM allows visualization of image regions influencing predictions, enhancing clinician trust and model interpretability.

Future work could explore multimodal learning approaches to further improve diagnostic accuracy. By combining image-based features with patient metadata—such as age, comorbidities, wound duration, and prior treatments—the model could deliver more personalized and context-aware assessments, bridging the gap between visual analysis and clinical context. Such advancements would strengthen decision-making and improve patient outcomes.

CHAPTER 5 – CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

This study successfully demonstrated the potential of Convolutional Neural Networks (CNNs) in automating multi-class wound classification from medical images. By utilizing a dataset of 8,652 images across 14 wound categories, the proposed deep learning framework integrated effective preprocessing, augmentation, and optimization techniques, achieving a validation accuracy of 84.22%. The results highlighted the model's strong predictive capability for certain wound types such as venous and pressure injuries, although performance varied for visually similar categories like burns and cuts due to dataset limitations and class imbalance. Despite these challenges, the study underscores the capability of CNN-based systems to assist in wound assessment, severity grading, and telemedicine-driven clinical decision-making. Moreover, with the integration of Groq SDK, the model demonstrated feasibility for real-time, interpretable AI inference providing a scalable solution for healthcare applications that require speed, reliability, and transparency.

5.2 FUTURE ENHANCEMENTS

Future work will aim to address current limitations and further strengthen the system's reliability, accuracy, and clinical value. Expanding the dataset with more diverse and balanced wound images will be prioritized to reduce class imbalance and improve generalization across different wound types. Additionally, the integration of **explainable AI (XAI)** techniques, such as **Grad-CAM** and **SHAP**, will enhance interpretability by providing clear visual explanations of the model's predictions—critical for gaining clinician trust on **real-world clinical validation**, deployment within **telemedicine platforms**, and **optimization via Groq SDK** to achieve real-time inference, ensuring the system is both medically reliable and practically deployable in clinical environments.

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APPENDICES

WORK CONTRIBUTION

Member 1 – TRIDEV BALAJI R

He studied recent research on Convolutional Neural Networks (CNN) and interpretable AI in dermatology and wound analysis to understand current advancements and challenges. The problem was defined as developing an interpretable AI backend capable of predicting wound conditions and offering actionable healing advice without requiring user login. I explored the capabilities of the Groq SDK, evaluating its performance benefits and suitability for accelerating AI model inference. During the process, I identified key challenges related to data quality, preprocessing, model interpretability, and the integration of language models for generating reliable medical responses. After discussing the approach with my supervisor, I received valuable guidance on utilizing medical-grade datasets, enhancing explainability, planning deployment strategies, and building user trust in the system.

Member 2 – RAKSITH S S

He focused on establishing the conceptual foundation and implementation plan of the project, selecting Convolutional Neural Networks (CNN) as the primary technique for feature extraction and classification. I analyzed the CNN structure, its computational flow, and optimization strategies to ensure high accuracy and reliability. Additionally, I evaluated various CNN configurations to determine the most adaptable architecture for the project's objectives. Finally, I designed a scalable and future-ready implementation plan that supports integration with larger datasets and allows for the inclusion of extended features

in future developments.

Member 3 – SONIKA N

She developed a fully responsive frontend dashboard for managing skin disease images using React JS and Tailwind CSS to ensure a modular, clean, and modern design. The dashboard supports seamless image uploads through both drag-and-drop and manual selection, featuring real-time previews and validation for enhanced usability. It includes functionalities to label, categorize, and filter images for efficient organization and streamlined workflow. Furthermore, I integrated a secure backend and cloud storage solution to ensure data protection and scalability, laying the groundwork for future AI-based analysis and telemedicine applications.

Member 4 – DHARNISH P

He designed the input acquisition system to receive wound type and condition predictions from the CNN model, mapping these outputs to corresponding precaution rules through a rule-based system or the Groq SDK. To enhance usability, I implemented dynamic generation of user-friendly precaution text using templates or integrated language models. Additionally, I incorporated CNN explainability techniques such as Grad-CAM and SHAP to provide visual insights into model predictions, improving user understanding and trust. The final design included a structured precaution page that seamlessly combined classification results, visual explanations, and descriptive text for clear and effective output delivery.

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